

Proposed Datasets and Modelling Plan

For the Capstone Project

Explainable Road and Vehicle Safety Intelligence Using Structured Crash Data and Consumer Complaint Narratives

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Project purpose

This project proposes two related but separate vehicle-safety analytics modules. The first will use structured police-reported crash data to identify characteristics associated with injury-producing crashes. The second will use consumer complaint narratives to recommend an affected vehicle-component category for human review. The modules will not be joined or combined into one score because they describe different events and populations.

Objectives

- Develop an explainable model for distinguishing injury or fatality crashes from property-damage-only crashes.
- Develop an interpretable text-classification model for routing vehicle-safety complaint narratives.
- Compare justified models, imbalance treatments and evaluation measures using chronological partitions.
- Demonstrate the selected pipelines as separate human-supervised modules in a local application.

Proposed datasets

Dataset	Description	Proposed use
NHTSA Crash Report Sampling System	Real structured relational crash data covering 2020-2024. Five verified annual archives contain 28 CSV tables per year.	Binary crash-severity analysis
NHTSA Vehicle Owner Complaints	Real structured metadata and unstructured narratives. The verified file contains 2,250,304 records and 51 documented fields.	Affected-component complaint routing

All raw datasets, the CRSS Analytical User's Manual, the 2020-2024 Coding and Validation Manuals, the complaint field layout and the complaint import guide have been downloaded and checked.

Dataset 1 Crash Report Sampling System

CRSS is maintained by the United States National Highway Traffic Safety Administration and is a nationally representative probability sample of police-reported traffic crashes. The five annual archives provide crash, vehicle, person, roadway and event information. Each field will be interpreted using the official analytical and annual coding manuals. The 2024 extraction contains approximately 676 MB; the exact combined extracted size of all five years will be measured during Data Understanding.

The principal tables use different units of observation. In the verified 2024 files, the accident table contains 51,658 sampled crashes, the vehicle table contains 90,641 involved vehicles, and the person table contains 126,159 involved persons. Equivalent annual structures will be checked and harmonised across 2020–2024 using the official manuals.

Preparation and target

The principal accident, vehicle and person tables will be linked with the documented keys CASENUM, VEH_NO and PER_NO. Vehicle and person information will be aggregated to crash level without duplicating crashes. The proposed binary target is injury or fatality crash versus property-damage-only crash. Identifiers and fields that directly encode injury severity, injury counts or post-crash outcomes will be excluded from predictors to prevent leakage. Unknown and not-applicable codes will be handled according to the manuals, and sampling weights will be used for population-level descriptive estimates.

Modelling and evaluation

A dummy classifier and Logistic Regression will establish baselines. Decision Tree, Random Forest and one boosted-tree method such as Histogram Gradient Boosting or XGBoost will then be compared. CRSS 2020–2022 will be used for training, 2023 for validation and 2024 as the untouched test year. Class weighting and training-only random undersampling will be compared if the observed target distribution warrants treatment. Evaluation will include ROC-AUC, PR-AUC, precision, recall, F1, confusion matrices, calibration and validation-to-test stability.

Dataset 2 NHTSA Vehicle Owner Complaints

The complaint dataset contains vehicle-safety reports submitted to NHTSA since January 1, 1995. It is a headerless, tab-delimited file, but the official layout provides the names, order, types and meanings of all 51 fields. The verified file contains 2,250,304 records, of which 2,250,265 contain narratives and 2,250,155 contain component descriptions. Each row represents a complaint-related product or component record. Consequently, one complaint reference may appear in multiple rows when several components are recorded. CDESCR contains the narrative and will be the primary text input. COMPDESC provides the proposed affected-component target and will be excluded from predictors. ODINO will keep related records from one complaint in the same data partition. Identifiers will be retained only for traceability and duplicate control

NLP modelling and evaluation

Training-fitted TF-IDF word unigrams and bigrams will represent the narratives. A majority-class baseline, Multinomial Naive Bayes, Logistic Regression and Linear Support Vector Classification will be compared. Records will be divided chronologically and exact or near-duplicate narratives will be audited before splitting. Evaluation will include macro-F1, weighted-F1, per-class precision and recall, confusion matrices and performance for both common and uncommon component categories.

CRISP DM methodology

Phase	Application
Business Understanding	Define users, decisions, restrictions and measurable success criteria.
Data Understanding	Document provenance and fields; examine quality, joins, missingness, duplicates, imbalance, time patterns and leakage.
Data Preparation	Harmonise years, decode values, create chronological partitions and fit transformations using training data only.
Modelling	Train baselines and candidate models; tune with training and validation data only.
Evaluation	Use untouched test data, class-level metrics, calibration, error analysis and objective-based interpretation.
Deployment	Save verified pipelines and expose two separate modules through an API and Streamlit interface.

Proposed demonstration

The crash module will accept the documented inputs required by the selected structured pipeline and return a severity score or classification, threshold, important factors and missing-input warnings. The complaint module will accept a narrative and return the recommended component category, confidence and influential terms. Both modules will state that their outputs support human review and do not prove causation, liability or the existence of a defect.

Main limitations

- CRSS is a probability sample, so weighted national estimates must be distinguished from raw sample counts.
- Codes may change between years and must be harmonised with the annual manuals.
- Complaint reports are voluntary allegations and do not independently verify defects or causation.
- Complaint counts are not defect rates because a vehicle-exposure denominator is unavailable.
- The datasets have no valid record-level linkage and their scores will remain separate.

Approval requested

Approval is requested to proceed with CRSS 2020-2024 as the structured component and NHTSA Vehicle Owner Complaints as the unstructured component. Together they provide real, documented, non-synthetic structured and unstructured data at sufficient scale for the capstone. If approved, development will follow all CRISP-DM phases with field-level explainability, leakage control, chronological evaluation and human oversight.

Official sources

- NHTSA Crash Report Sampling System: <https://www.nhtsa.gov/crash-data-systems/crash-report-sampling-system>
- NHTSA Datasets and APIs: <https://www.nhtsa.gov/nhtsa-datasets-and-apis>
- NHTSA CRSS downloads: <https://www.nhtsa.gov/file-downloads?p=nhtsa/downloads/CRSS/>